Show cars whose 7-day average temperature rose more than 10F versus their 30-day baseline.

Thought process ›

Cars with 7-day average temperature up more than 10°F vs their 30-day baseline

74 cars currently show a 7-day average temperature at least 10°F above their 30-day baseline average, indicating a recent heat-up trend that could warrant inspection attention 1.

The largest jumps are concentrated in a few outliers: **NS100392** (Boxcar, Pittsburgh) is up 40.7°F (115.2°F vs 74.5°F baseline), **NS100632** (Tank, Harrisburg) is up 36.3°F, **NS100356** (Hopper, Alabama) is up 33.5°F, and **NS100292** (Gondola, Atlanta) is up 33.4°F. Most of the remaining cars cluster in the 11–20°F rise range, spanning all five divisions (Pittsburgh, Harrisburg, Atlanta, Chicago, Alabama) and all car types (Boxcar, Tank, Hopper, Gondola, Intermodal).

Show code

Cars with 7d avg temp >10F above 30d baseline

Car ID	Car Type	Division	Age (yrs)	Avg Temp 7d (F)	Avg Temp 30d (F)	Temp Rise (F)
NS101033	Boxcar	Pittsburgh	5	58.7	43.9	14.8
NS100639	Gondola	Alabama	34	60.2	45.6	14.6
NS100348	Hopper	Chicago	8	52.4	37.8	14.6
NS101442	Gondola	Alabama	18	50.9	36.5	14.4
NS101258	Tank	Chicago	23	51.6	37.6	14
NS101629	Hopper	Pittsburgh	33	59.4	45.5	13.9
NS100136	Intermodal	Alabama	22	55	41.5	13.5

< 1 2 3 >



Source: Full critical risk car list

1 source

How many set-outs were false alarms, by division?

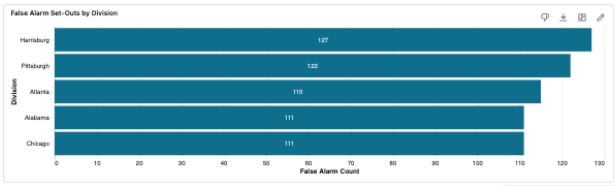
Thought process

False Alarm Set-Outs by Division

False-alarm set-outs are fairly evenly distributed across all five divisions, totaling **586 false alarms** overall [1].

- Harrisburg: 127
- Pittsburgh: 122
- Atlanta: 115
- Alabama: 111
- Chicago: 111

Show code



Source: False alarm set-outs by division

No single division stands out as a major outlier — counts range narrowly from 111 to 127, roughly a 14% spread between the highest (Harrisburg) and lowest (Alabama/Chicago, tied).

1 source

What share of set-outs were confirmed defects?

Thought process

Confirmed Defect Share of Set-Outs

Of the 940 total set-outs, **354 (37.7%)** were confirmed defects, while the remaining 586 (62.3%) were false alarms [1].

1 source

Show the 5 highest-risk cars in Pittsburgh with their 7-day average temperature and trend.

Thought process

Top 5 Highest-Risk Cars in Pittsburgh

Three of the top five Pittsburgh cars carry the maximum risk score of 1.0 — **NS101583** (Tank), **NS101999** (Boxcar), and **NS101492** (Hopper) — all showing elevated 7-day average temperatures (89–95°F) and strong upward trend deltas (16.5–24.9°F) [1].

Show code

Top 5 Highest-Risk Cars in Pittsburgh

Car ID	Car Type	Risk Score	Avg 7d Temp (F)	Trend Delta (F)	Age (yrs)
NS101583	Tank	1	89.1	16.5	12
NS101999	Boxcar	1	95	21.6	21
NS101492	Hopper	1	78.7	24.9	16
NS100392	Boxcar	0.807	57.4	6.3	25
NS101902	Intermodal	0.044	66.2	17.6	20

Source: Top 5 highest risk cars in Pittsburgh

Risk drops off sharply after the top three: **NS100392** sits at 0.81 risk with a modest 6.3°F trend, and **NS101902** rounds out the top five at just 0.04 risk despite a 17.6°F trend rise — its lower baseline temperature keeps it below the critical threshold.

1 source

Which divisions have cars with risk above 0.5 and a temperature trend rising more than 10 degrees?

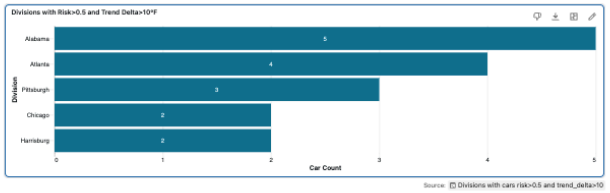
Thought process

Divisions with High-Risk, Rising-Temperature Cars

All five divisions have at least one car with a risk score above 0.5 and a temperature trend rising more than 10°F, for a total of 16 qualifying cars [1].

- Alabama: 5
- Atlanta: 4
- Pittsburgh: 3
- Chicago: 2
- Harrisburg: 2

Show code



Source: Divisions with cars risk>0.5 and trend_delta>10

Alabama leads with the most cars meeting both criteria, followed by Atlanta — these two divisions account for over half (9 of 16) of all qualifying cars.